Bulk RNA-seq: from reads to differential expression

Theory, tools, pros and cons

1 QC
FastQC, fastp, MultiQC

2 Align or pseudo-align
STAR, HISAT2, Salmon, kallisto

3 Count
featureCounts, tximport

4 Test
DESeq2, edgeR, limma-voom

5 Interpret
GSEA, clusterProfiler

Statistical theory

Count model

Reads per gene are counts with more variance than Poisson, so they are modelled with a negative binomial distribution (mean and dispersion).

Normalization

Library size and composition are corrected with median-of-ratios (DESeq2) or TMM (edgeR). TPM and FPKM are not for testing.

Few replicates

Dispersion is shared across genes and shrunk toward a trend (empirical Bayes), which stabilizes small-n estimates.

Many tests

Thousands of genes are tested at once, so p-values are adjusted for FDR (Benjamini-Hochberg).

Main differential expression tools

Tool	Model	Strength	Caution
DESeq2	NB GLM, shrunken LFC	Robust defaults, good small-n behavior	Can be conservative; outlier-sensitive
edgeR	NB GLM, quasi-likelihood	Flexible designs, fast	Results sensitive to filtering
limma-voom	Linear model on log-CPM with weights	Fast, scales to large cohorts, complex designs	Needs enough replicates for weights

Benchmarks disagree: Baik et al. 2020 found performance depends on the benchmark used; DESeq2, robust edgeR and voom-TMM did well overall.

Bulk RNA-seq: pros and cons

Pros

- Genome-wide, unbiased, no probes needed
- Wide dynamic range, good for low-expressed genes
- Detects isoforms, fusions and novel transcripts
- Cheaper than single-cell, mature tools
- Acquired to show heterogeneity of transcripts
- Gives good representation of the biggest populations
- Deconvolution of samples can give attribution of different cell populations per sample

Cons

- Averages over cell types, hiding heterogeneity
- Relative, not absolute, abundance
- Sensitive to batch, RNA quality and depth
- Short reads limit isoform resolution
- Sensitive to sampling or bias in library composition/amplifcation
- Lack of spatial information as well for each sample

Common problems and fixes

Problem	What to do
Too few replicates	Plan at least 6 per group (Schurch et al. 2016)
Batch effects	Randomize, include batch in the design
Low-count genes and outliers	Filter consistently, inspect PCA & hierarchical clustering
Tools give different gene lists	Run two methods, report overlap, and inspect scored genes

Key papers: Conesa 2016; Love 2014 (DESeq2); Robinson 2010 (edgeR); Law 2014 (voom); Baik 2020.